Footprinting with theHarvester: microsoft.com

Nov 9, 2026 · @sheila

theHarvester found 3 email addresses, 868 hosts and 147 IPs for microsoft.com using all sources, while the Baidu-only run found nothing.

Objective and scope

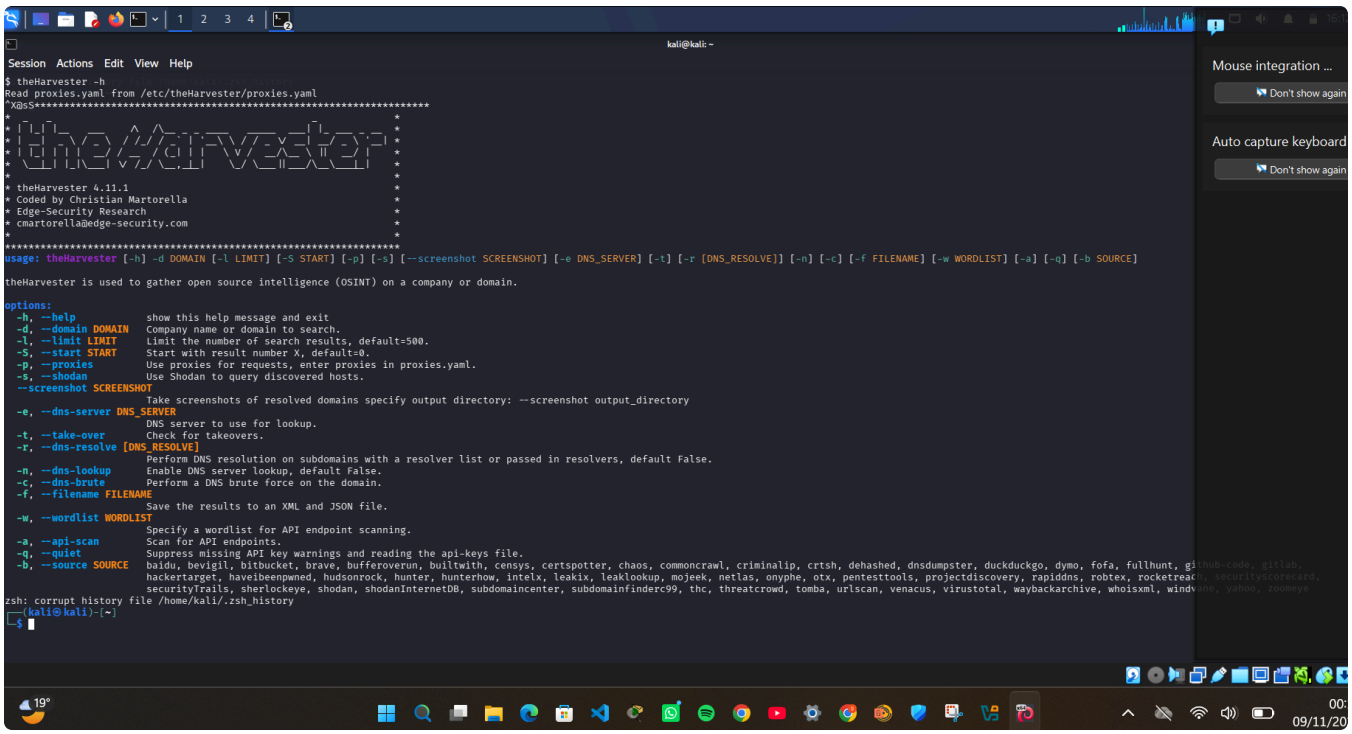
The task is to gather email addresses and sub-domains for microsoft.com, the target assigned in project module W2-PM4. theHarvester 4.11.1 was run on Kali Linux. It only queries public sources, so nothing on the target was accessed directly.

Task 1: Baidu, limit 1000

The usage help was read first (Figure 1). The search was then run with Baidu as the only source, a limit of 1000 results, and the output saved to task1_baidu.txt:

```
theHarvester -d microsoft.com -l 1000 -b baidu | tee task1_baidu.txt
```

Here -d is the target domain, -l limits the results and -b sets the data source. The run completed without errors and found no IPs, emails, people or hosts (Figure 2). The saved file is 1075 bytes and holds the same output (Figure 3).


kali@kali: ~
Session Actions Edit View Help
$ theHarvester -h
Read proxies.yaml from /etc/theHarvester/proxies.yaml
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-S START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c] [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]
theHarvester is used to gather open source intelligence (OSINT) on a company or domain.
options:
-h, --help show this help message and exit
-d, --domain DOMAIN Company name or domain to search.
-l, --limit LIMIT Limit the number of search results, default=500.
-S, --start START Start with result number X, default=0.
-p, --proxies Use proxies for requests, enter proxies in proxies.yaml.
-s, --shodan Use Shodan to query discovered hosts.
--screenshot SCREENSHOT
Take screenshots of resolved domains specify output directory: --screenshot output_directory
-e, --dns-server DNS_SERVER
DNS server to use for lookup.
-t, --take-over Check for takeovers.
-r, --dns-resolve [DNS_RESOLVE]
Perform DNS resolution on subdomains with a resolver list or passed in resolvers, default False.
-n, --dns-lookup Enable DNS server lookup, default False.
-c, --dns-brute Perform a DNS brute force on the domain.
-f, --filename FILENAME
Save the results to an XML and JSON file.
-w, --wordlist WORDLIST
Specify a wordlist for API endpoint scanning.
-a, --api-scan Scan for API endpoints.
-q, --quiet Suppress missing API key warnings and reading the api-keys file.
-b, --source SOURCE baidu, bevigil, bitbucket, brave, bufferoverun, builtwith, censys, certspotter, chaos, commoncrawl, criminalip, crtsh, dehashed, dnsdumpster, duckduckgo, dymo, fofa, fullhunt, github-code, gitlab,
hackertarget, haveibeenpwned, hudsonrock, hunter, hunterhow, intelx, leakix, leaklookup, mojeek, netlas, onyphe, otx, pentesttools, projectdiscovery, rapiddns, robtex, rocketreach, securityscorecard,
securityTrails, sherlockeye, shodan, shodanInternetDB, subdomaincenter, subdomainfinderc99, thc, threatcrowd, tomba, urlscan, venacus, virustotal, waybackarchive, whoisxml, windvane, yahoo, zoomeye
zsh: corrupt history file /home/kali/.zsh_history
Mouse integration ...
Don't show again
Auto capture keyboard
Don't show again


zsh: corrupt history file /home/kali/.zsh_history
rm ~/.zsh_history
theHarvester -d microsoft.com -l 1000 -b baidu | tee task1_baidu.txt
Read proxies.yaml from /etc/theHarvester/proxies.yaml
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
[*] Target: microsoft.com
[*] Searching Baidu.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] No hosts found.


ls -l task1_baidu.txt
-rw-rw-r-- 1 kali kali 1075 Nov 8 16:14 task1_baidu.txt
cat task1_baidu.txt
Read proxies.yaml from /etc/theHarvester/proxies.yaml
* theHarvester 4.11.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
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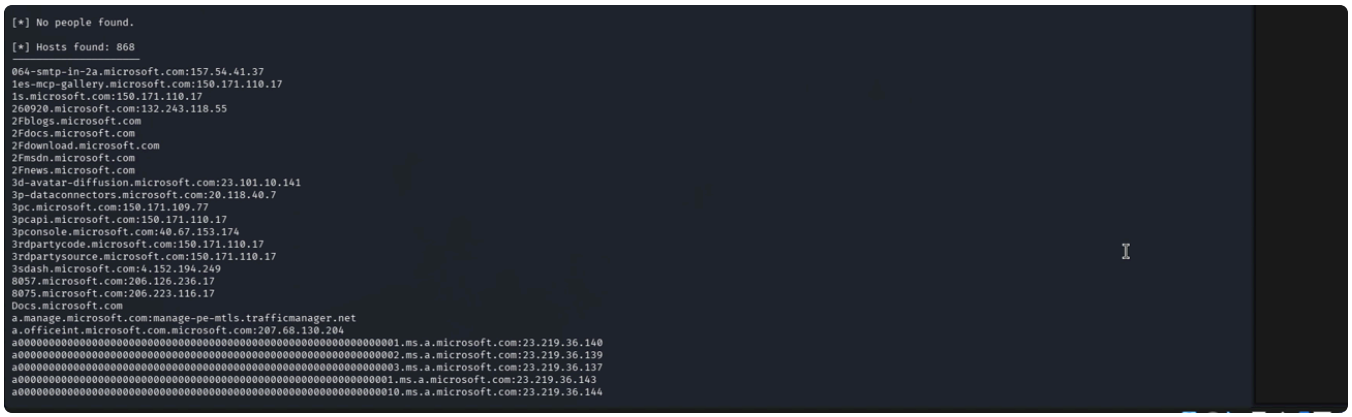
Task 2: all sources, limit 50

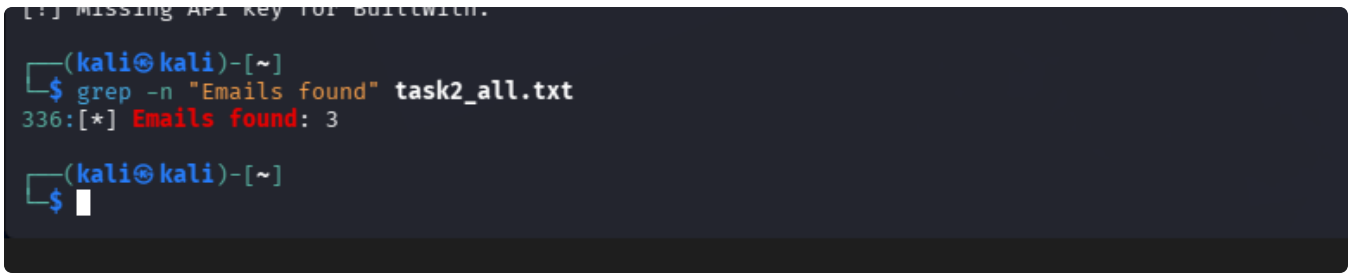
The search was repeated with all sources and a limit of 50, saving the output to task2_all.txt:

```
theHarvester -d microsoft.com -l 50 -b all | tee task2_all.txt
```

The run took several minutes. Many sources reported a missing API key (Figure 4), and the tool carried on with the sources that did not need one, including Baidu, Certspotter, crt.sh, Commoncrawl, DuckDuckGo, Hackertarget, Hudson Rock, Otx, Rapiddns, Mojeek and Robtex. The summary reports 868 hosts (Figure 5), and a search of the saved file confirms the emails count (Figure 6).







Findings

Run	Source	Limit	Emails	Hosts	IPs
Task 1	Baidu	1000	0	0	0
Task 2	All sources	50	3	868	147

The three emails from Task 2 are dotnet-docker-bot@microsoft.com, opencode@microsoft.com and secure@microsoft.com. They look like automated or role-based mailboxes, not named staff. No people and no LinkedIn links were found. The 868 hosts are sub-domains of microsoft.com, for example account, careers, cdn, dotnet and download.

Limitations

The results differ from the project's sample, which found one email with Baidu. theHarvester depends on public sources that change over time, and the project notes that results may vary. Baidu returned nothing in this run. Many sources need API keys that were not configured, so the Task 2 results are likely smaller than a fully keyed run. The run also saw authentication errors from Fofa and LeakIX, and Shodan InternetDB could not resolve the domain.

Risk and mitigation

Harvested emails can feed phishing and password attacks, and each sub-domain is another system an attacker can probe, including forgotten ones. To reduce exposure, keep an inventory of sub-domains and retire unused ones, use role-based mailboxes behind email filtering, and run theHarvester against your own domain regularly to see what an attacker would see.